

ZIMBABWE

MINISTRY OF PRIMARY AND SECONDARY EDUCATION

HORTICULTURE SYLLABUS

FORMS 5 - 6

2024 - 2030

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1.0 PREAMBLE

1.1 INTRODUCTION

Zimbabwe embarked on an agrarian Land Reform to correct colonial land imbalances and empower the previously marginalized Zimbabweans with access to land. In-or- der to safe guard this important national heritage, ensure food security through sustainable land use, it is import- ant for learners of diverse backgrounds to acquire necessary horticultural knowledge and skills. Horticulture is the art and science of producing fruits, vegetables, flowers, herbs, ornamental plants and also involves landscaping. This Heritage-Based Horticulture syllabus is designed for Form 5 and 6 learners in Horticulture. It is a two-year learning phase which covers concepts, principles and practices in horticulture. The syllabus will provide learners with a rich experience in identifying, investigating, problem solving and assessing the viability of sustainable horticultural enterprises in an indigenized economy. Learners will be assessed through continuous and summative assessment.

1.2 RATIONALE

Agriculture is a learning area studied from Grade 3 to Form 4 therefore it is imperative for learners to specialize at Form 5 and 6 so as to acquire adequate horticultural skills and knowledge to create employment and for further learning opportunities. Specialization in horticulture would enable learners to be proactive, productive, add value to the community and national economy.

Horticulture stimulates in learners, the responsibility to intensively care and utilize land sustainably for economic development. The Horticulture learners will at the end of the two –year learning phase, value the dignity of labour and food sovereignty.

- Enterprise development
- Designing skills

1.3 SUMMARY OF CONTENT

This Heritage-Based Horticulture learning area will include the study of background to horticulture, production technology, plant physiology, soil and water management, plant protection, landscaping and ornamental plants, floriculture, vegetable production, fruit production and herb production

1.4 ASSUMPTIONS

It is assumed that learners have practical skills and knowledge in:

- Crop production
- horticulture production
- sustainable use of agricultural resources
- e-learning
- marketing of horticultural products

1.5 CROSS- CUTTING ISSUES

The This Heritage-Based Horticulture learning area will encompass the following cross cutting themes:

- Disaster and risk reduction
- Enterprise skills
- Environmental issues
- Team work
- Sustainable resource utilization
- Digital literacy
- Inclusivity
- Safety and health
- Gender

2.0 PRESENTATION OF SYLLABUS

This Heritage-Based Horticulture syllabus is a single document covering Forms 5- 6. The syllabus has a suggested list of resources to be used during teaching and learning.

3.0 AIMS

The Heritage-Based Horticulture syllabus aims to help learners to:

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3. 1 develop an appreciation of the socio-economic importance of horticulture to agricultural development of the country

3.2 develop positive attitudes towards Horticulture as a learning area

3.3 apply competences in solving horticulture related problems

3.4 demonstrate innovativeness in the sustainable utilization of local resources in intensive horticulture production

3.5 apply value addition skills in the processing and marketing of horticultural products to meet food security and economic standards

3.6 demonstrate desirable literacy and numeracy skills including practical competences necessary for life

3.7 prepare for life and work in an indigenized economy, increasingly globalised and competitive environment

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4.0 SYLLABUS OBJECTIVES

By the end of the learning phase learners should be able to:

4.1 demonstrate an understanding of the importance of horticulture in socio-economic development
4.2 apply scientific principles in horticultural production
4.3 evaluate resources necessary for horticulture production
4.4 demonstrate an understanding of plant protection principles
4.5 demonstrate the sustainable utilization of local resources
4.6 efficiently produce and market horticultural products
4.7 add value to horticultural products
4.8 design and carry out research work on horticulture production for economic development of the nation
4.9 prepare and implement a sustainable horticultural project proposal

5.0 METHODOLOGY AND TIME ALLOCATION

5.1 METHODOLOGY

Learner centred and hands on approaches should be used in the development of concepts and skills. These approaches should be inclusive and encourage curiosity as well as promote practical oriented learning. Emphasis should be placed on equipping learners with practical skills. Linkage between theory and practice should be implemented in the teaching and learning of horticulture

The following are suggested methods of teaching and learning of horticulture:

- Discussions
- Demonstrations
- Experimentations
- Problem-solving
- E – learning
- Debates
- Specimen collections
- Research
- Project-based learning
- Educational tours
- Design-based learning
- Survey
- Simulation and modelling

6.0 TOPICS

1. Background to Horticulture
2. Production technology
3. Plant physiology
4. Soil and water management
5. Plant protection
6. Landscaping and ornamental plants
7. Floriculture
8. Vegetable Production
9. Fruit Production
10. Herb and Spice production

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7.0 SCOPE AND SEQUENCE

7.1 TOPIC 1: BACKGROUND TO HORTICULTURE

TOPIC	FORM 5	FORM 6
Background to Horticulture	• Historical perspective • Branches • Importance • Factors affecting horticulture production	

7.2 TOPIC 2: PRODUCTION TECHNOLOGY

TOPIC	FORM 5	FORM 6
Production technology	• Production systems • Propagation methods • Structures and Equipment used in horticulture production	

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7.3 TOPIC 3: PLANT PHYSIOLOGY

TOPIC	FORM 5	FORM 6
Plant structure	• Roots • Stems • Leaves • Flowers	
Plant-water relations	• Water properties • Water movement • Radial movement of water • Water potential • Transpiration •	
Bioenergetics and ATP synthesis	• Photosynthesis • Photosynthetic pathways • Translocation • Cellular respiration	
Plant Growth and Development	• Germination • Meristems • Plant growth • Plant growth regulators	
Environmental factors	• Effects of environmental factors	

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7.4 TOPIC 4: SOIL AND WATER MANAGEMENT

TOPIC	FORM 5	FORM 6
Soil	• Physical properties • Chemical properties • Biological properties • Soil management	
	• Soil moisture • Water and the environment • Soil-water management	

7.5 TOPIC 5: PLANT PROTECTION

Plant protection	• Weeds • Pests • Diseases • Safety precautions • Sprayer calibration	

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7.6 TOPIC 6: LANDSCAPING AND ORNAMENTAL PLANTS

TOPIC	FORM 5	FORM 6
Landscaping	• Landscaping • Principles of landscaping • Designing	
Ornamental plants	• ornamental plants	

7.7 TOPIC 7: FLORICULTURE

TOPIC	FORM 5	FORM 6
Flower production	• Origin and uses of flowers • Land preparation • Management practices Post-harvest handling and marketing	• Flower production

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7.8 TOPIC 8: VEGETABLE PRODUCTION

TOPIC	FORM 5	FORM 6
Vegetable production	• Vegetables • Vegetable rotation • Environmental requirements • Nursery • Vegetable management • Harvesting • Post-harvest handling and marketing	• Environmental requirements • Nursery • Vegetable management • Harvesting • Post-harvest handling and marketing

7.9 TOPIC 9: FRUIT PRODUCTION

TOPIC	FORM 5	FORM 6
Fruit production		• Importance of fruits • Classification • Nursery • Propagation • Orchard establishment • Orchard management • Harvesting • Post-harvest handling and marketing

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7.10 TOPIC 10: HERB AND SPICE PRODUCTION

TOPIC	FORM 5	FORM 6
Herb production		• Herb production
Spice production		• Spice production

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8.0 COMPETENCY MATRIX

FORM 5 SYLLABUSES

8.1 TOPIC 1: BACKGROUND TO HORTICULTURE

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Historical perspective	• discuss the origin of horticulture • explain the development of horticulture	• Origin of horticulture • Development of horticulture locally, regionally and globally	• Discussing the origin of horticulture • Explaining the development of horticulture • Researching on the origins of horticulture	• ICT tools with Jaws software • Textbooks
Branches of horticulture	• outline the branches of horticulture • describe the branches	• Vegetable production • Flower production • Fruit production • Herb production	• Identifying the branches of horticulture • Discussing the branches of horticulture • Educational touring horticulture farms	• ICT tools with Jaws software • Horticulturalist
Importance of horticulture	• discuss the socio-economic importance of horticulture • explain the ecological importance of horticulture	• Socio-economic importance • Ecological importance	• Discussing the socio-economic importance of horticulture • Explaining the ecological importance of horticulture • Researching on socio-economic importance of horticulture	• ICT tools with Jaws software • Horticulturalist
Factors affecting horticulture production	• identify factors affecting horticulture production • discuss how each factor affects horticulture production	• Environmental factors • Economic factors • Cultural factors • Religious factors • Political factors	• Discussing the factors affecting horticulture production • Experimenting on how environmental factors affect horticulture production • Assessing the impact of cultural, political and religious	• ICT tools with JAWS software

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• discuss mitigatory measures against negative effects		factors on horticulture production • Generating mitigatory measures against negative effects	

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8.2 TOPIC 2: PRODUCTION TECHNOLOGY

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Production systems	• identify production systems in horticulture • describe the nature and function of controlled environments • discuss the strengths and weaknesses of each production system	• Open field • Green house • Shade house • Hot bed • Cold frames	• Discussing production systems of horticulture crops • Designing controlled environments • Evaluating the strengths and weaknesses of each production systems • Constructing a shade house and demonstrating its use • Educational touring to a greenhouse production system	• ICT tools with JAWS software • Print and electronic media • Green house • Shade house • Hot bed • Cold frames
Propagation methods	• discuss propagation methods in horticulture • illustrate how to propagate using seeds, layering, cuttings, grafting and budding • describe the concepts of cloning and tissue culture • outline practical applications of the tissue culture technique	• Seed • Suckers • Cuttings • Grafting and budding • Layering • Tissue culture • Cloning	• Discussing propagation methods in horticulture • Demonstrating how to propagate different plants using seeds, cuttings, grafting, budding and layering • Illustrating the concepts of cloning and tissue culture • Viewing video clips on tissue culture techniques in production of horticultural plants • Raising horticulture plants using vegetative propagation methods	• ICT tools with JAWS Software • Print and electronic media • Seed • Vegetative propagules • Tissue

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Structures and equipment used in Horticulture	• discuss the structures used in horticulture production • relate the design of the structures to their functions • demonstrate how to use horticultural equipment	• Driers • Storage facilities • Equipment used in horticulture	• Describing structures used in horticulture production • Demonstrating how to use horticultural equipment	• ICT tools with JAWS Software • Print and electronic media • Horticultural tools and equipment

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8.3 TOPIC 3: PLANT PHYSIOLOGY

SUB TOPIC: PLANT STRUCTURE

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Roots	• identify parts of the cross-section of a root • outline functions of root parts • prepare root slides	• Root anatomy	• Cutting a root to view the cross-section • Outlining functions of root parts • Preparing slides • Viewing slides • Drawing of the cross-section of a root	• Roots, stems • ICT tools with JAWS Software • Slide strips • Microscope • Prepared slides
Stems	• identify parts of the cross-section of a stem • outline functions of the stem parts • prepare stem slides	• Stem anatomy	• Cutting a plant stem • Outlining functions of stem parts • Drawing of the cross-section of stem • Preparing stem slides	• stems • ICT tools with JAWS Software • Slide strips • Microscope • Prepared slides
Leaves	• Identify parts of the cross-section of a leaf • Outline functions of leaf parts • Prepare leaf slides	• Leaf anatomy	• Viewing slides • Drawing the cross-section of a leaf • Labeling parts of the cross-section of a leaf • Outlining functions of leaf parts • Preparing leaf slides	• Leaf samples • ICT tools • Slide strips • microscope
Flowers	• Identify parts of the cross-section of a flower	• Flower anatomy - Wind	• Viewing slides • Drawing of the cross -ection of	• Flower samples • ICT tools

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• Outline functions of flower parts • Differentiate wind from insect pollinated flower • Distinguish between a male and female flower	- Insect pollinated	a flower • Identifying parts of the cross-section of a flower • Describing functions of flower parts • Distinguishing wind from insect pollinated flowers	• Slide strips • Microscope

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SUBTOPIC: PLANT-WATER RELATIONS

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Water properties	• Describe water properties in relation to its functions	• Water properties: - cohesion - adhesion - universal solvent - specific heat capacity - heat of fusion - heat of vaporization	• Discussing water properties • Conducting experiments on water properties	• ICT tools • Water samples
Water movement	• discuss factors that affect water uptake in plants • explain the mechanism of water uptake in plants	• Factors affecting water uptake • Osmosis • Diffusion • Mass flow	• Discussing factors that affect water uptake • explaining the mechanism of water uptake • carrying out experiments to demonstrate water movement	• ICT tools • Potato cubes • Experiment kit
Radial movement of water	• describe water flow pathways from cell to cell	• Pathways of water movement: - apoplast - symplast. - vacuolar	• Discussing water flow pathways from cell to cell. • Illustrating water flow path ways from cell to cell	• ICT tools with JAWS software

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Water potential	• explain the components of water potential • design and carry out experiments on water potential • determine water potential	• Water potential - pressure potential - osmotic potential - matric potential	• Discussing the components of water potential • Designing and carrying out experiments on water potential • Calculating water potential	• ICT tools with JAWS software • Potato cubes • Experimental kit
Transpiration	• describe factors affecting the rate of transpiration	• Factors affecting transpiration: - light - temperature - humidity - wind - soil water - plant factors	• Discussing factors affecting the rate of transpiration • Designing experiments to investigate the effects of factors on the rate of transpiration	• Experimental kits

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SUBTOPIC: BIOENERGETICS AND ATP SYNTHESIS

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Photosynthesis	• explain the factors affecting the rate of photosynthesis	• Factors affecting rate of photosynthesis - light intensity - carbon dioxide concentration - oxygen concentration - water - temperature	• Discussing the factors affecting rate of photosynthesis • Illustrating the effects of environmental factors on photosynthesis • Carrying out experiments on factors affecting photosynthesis	• ICT tools with JAWS software • Plant samples
	• describe the light-dependent reactions	• Light dependent reactions: - photolysis of water - cyclic and non-cyclic phosphorylation • Light-independent reactions	• Describing the light-dependent reactions	
	• explain the process of carboxylation • describe photosynthetic electron transport		• Describing the process of carboxylation	
	• describe bio-chemical mechanism by which ATP is synthesized • explain the universal role of ATP as the energy 'currency' in plants	• ATP synthesis	• Illustrating the structure and synthesis of ATP, • Demonstrating the bio-chemical mechanism by which ATP is synthesized • Describing the universal role of ATP as the energy 'currency' in plants	
Photosynthetic pathways	• describe the structural differences between C_3 and C_4 plants • compare the C_3 and C_4 bio	• Bio-chemical pathways - C_3	• Illustrating C_3, C_4 and CAM pathways • Discussing the structural	• ICT tools with JAWS software • Plant samples

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	chemical pathways	- C4 -Crassulacean Acid Metabolism (CAM)	differences between C3 andC4 plants • Differentiating C3, C4 and CAM bio-chemical pathways	• Print Media
Translocation	• relate the phloem structure to its functions • describe the translocation mechanism of different photosynthates • explain the significance of translocation in plants	• Phloem structure • Mechanisms: - Active - Mass flow - Diffusion	• Describing the structure of the phloem and relate it to its functions. • Discussing translocation mechanism of different photosynthates • Carrying out experiments to verify translocation in plants • Discussing the importance of translocation in plants	• ICT tools with JAWS software • Plant samples • Print Media
Cellular respiration	• outline the factors affecting respiration • describe the process of respiration. • discuss the significance of respiration in plants • apply the knowledge of cellular respiration in horticulture	• factors affecting cellular respiration • Glycolysis • Kreb's cycle • Electron transport chain • Significance of respiration in plants	• Discussing factors affecting respiration • describing, the process of respiration • discussing the significance of respiration in plants • Applying the knowledge of respiration in plants	• ICT tools • Fruits • Seeds • Plant samples

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SUB TOPIC: PLANT GROWTH AND DEVELOPMENT

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Seed germination	• discuss requirements for seed germination • describe the process of seed germination • distinguish between epigeal and hypogeal germination • test seed for viability • discuss the different types of seed dormancy • describe the methods of over-coming dormancy	• Requirements for optimum germination: - water - temperature - oxygen - light • Seed germination processes: - imbibition - enzyme activation • Types of seed germination: - epigeal - hypogea • Seed viability • Types of seed dormancy • Overcoming dormancy	Examining the requirements for seed germination • Describing the process of seed germination • Carrying out experiments on the conditions necessary for germination • Distinguishing between epigeal and hypogeal germination • Determining seed viability • Discussing types of seed dormancy • Carrying out experiments on over-coming seed dormancy	• ICT tools with JAWS software • Seed samples • Tetrazolium trichloride • Controlled environment
Meristems	• describe the types of plant meristems • apply the knowledge of meristems in horticultural crops	• Meristems: - apical - intercalary - lateral - basal	• Discussing types of plant merestems • Locating the meristematic sites on a plant • Discussing the importance of meristems in horticulture plants	• ICT tools with JAWS software • Meristems
Plant growth	• explain the concepts of growth and development • describe the phases of plant growth	• Growth and development: - primary and secondary	• Discussing growth and development in plants • Describing how cell division and enlargement lead to plant	• ICT tools with JAWS software • Plant samples

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	• differentiate determinant from indeterminant growth habits	- determinant and indeterminant growth	growth • Contrasting determinant and indeterminate growth habits	• Grafted plants
Plant Growth Regulators	• describe the effects of plant growth regulators on plant growth and development • apply the knowledge of plant growth regulators in horticulture	• Gibberillins • Cytokinins • Ethylene • Auxins • Abscisic acid	• Discussing effects of plant growth regulators on growth and development • Using plant growth regulators in horticulture production	• Plant growth regulators • Horticulturalist/Agronomist

SUB TOPIC: ENVIRONMENTAL FACTORS

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Effects of environmental factors	• explain the effects of environmental factors on horticultural production	• Environmental factors: - air circulation - drainage - light - humidity - temperature - moisture - soil fertility	• Discussing the effect of environmental factors on crop productivity. • Designing and carrying out experiments on responses of plants to environmental factors	• plant samples • ICT tools with JAWS software

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8.4 TOPIC 4: SOIL AND WATER MANAGEMENT

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Physical properties	• describe physical properties of soil • determine the physical properties of the soil • discuss the importance of soil physical properties to plant growth	• Soil texture • Soil color • Soil structure • Soil air • Bulk density • Particle density • Porosity	• Discussing the importance of soil physical properties • Determining the physical properties of soil • Relating bulk density to soil structure and porosity	• Soil samples • ICT tools with JAWS software
Chemical properties	• explain the effects of soil pH to crop production • correct soil pH for specific plants using suitable agents • explain the significance of cation exchange •	• Soil pH • Cation Exchange Capacity (CEC)	• Discussing the effects of soil pH on crop production • Determining soil pH • Applying pH correcting agents • Discussing the importance of cation exchange capacity •	• Soil samples • pH meters • ICT tools with JAWS softwares • Liming materials
Biological properties	• describe the importance of soil organisms in crop production • explain the factors that influence activities of soil organisms	• Soil macro-organisms: - earthworms - Termites - Ants • Soil micro - organisms: bacteria - Fungi - Nematodes - Protozoa	• Discussing the importance of soil organisms in crop production • Outlining the factors that influence activities of soil organisms • Carrying out experiments to determine presence of organisms in soil	• Soil samples • Plant samples • ICT tools with JAWS software
Soil management	• explain the roles of macro and micro elements in plants • apply the principles of soil	• Plant nutrients • Types of fertilize • Application methods • Calculations • Soil sampling and analysis	• Discussing the roles of macro and micro elements • Collecting soil samples • Analyzing soil results • Collecting plant samples	• Soil sampling tools • Fertilizer samples • Organic matter samples • Mulching materials • Digging tools

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	sampling in collecting soil samples • interpret soil analysis results • determine fertilizer requirements per given plant • differentiate the fertilizer types according to nutrients composition and formulation • select the appropriate application method • justify choice of appropriate soil conservation methods for specific areas • carry out appropriate soil conservation methods for an enterprise	• Soil conservation	showing deficiency symptoms • Calculating fertilizer requirements • Classifying the fertilizer types according to formulation and nutrient composition • Choosing the appropriate application methods • Identifying appropriate soil conservation methods • Implementing soil conservation measures	• ICT tools with JAWS software • Plant samples
Soil moisture	• discuss the importance of soil moisture • describe the forms of soil moisture • explain the terms used in soil water relations • determine soil moisture content • determine available soil moisture for plant uptake	• Importance of soil moisture • Capillary • Hygroscopic • Gravitational • Terms used in soil water relations: - field capacity - available water capacity - wilting point	• Discussing the importance of soil moisture • Outlining the forms of soil moisture • Describing the terms used in soil water relations • Designing and carrying out an experiment on soil moisture content • Calculating available soil moisture	• Soil moisture meter • Tension meters • Soil samples • ICT tools with JAWS software
Water and the environment	• discuss factors that influence soil moisture availability • manage soil moisture in horticulture production	• Factors affecting soil moisture availability - Soil texture - Organic matter content - Soil structure - Soil temperature - Transpiration - Soil salts	• Researching on factors influencing soil moisture availability • Carrying out experiments to investigate the effects of the environment on soil moisture availability	• Plants organic matter • Soil samples • ICT tools with JAWS software

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Soil-water management	• describe different irrigation systems which are suitable for horticulture • evaluate plant-water requirements using the principles of irrigation scheduling • describe various methods of conserving soil moisture	• Irrigation systems • Irrigation scheduling • Conservation methods	• Discussing different irrigation systems • Calculating plant- water requirements • Discussing various methods of conserving soil moisture • Touring horticulture enterprises	• ICT tools with JAWS software • Evaporation pan • Irrigation specialist • Irrigation equipment

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8.5 TOPIC 5: PLANT PROTECTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Weeds	• outline the socio-economic importance of weeds • identify weeds • classify weeds • describe mechanisms that make weeds persistent • explain the mechanisms of crop-weed competition • describe the principles of weed management practices	• Socio-economic importance of weeds. • Identification • Classification • Weed persistence mechanisms • Crop-weed competition: - inter and intra-specific competition • Principles of Weed Management: prevention eradication Integrated Weed Management (IWM)	• Discussing the socio-economic importance of weeds • Carrying out a project on weed identification and specimen preservation • Describing weed persistence mechanisms • Describing crop-weed competition • Discussing the principles of weed management	• Weeds • Herbicides • ICT tools
Pests	• outline the socio-economic importance of pests • identify pests • classify pests according to feeding habits • describe the life cycle of pests	• Socio-economic importance of pests: - insects - mites - nematodes • Identification of pests • Classification of pests according to feeding habits • Life cycle: - complete metamorphosis - incomplete metamorphosis	• Discussing the socio-economic importance of insect, mite and nematodes • Identifying and classifying pests according to feeding habits • Describing the life cycle of pests • Collecting and preserving local pest species	• Pest specimens • ICT tools with JAWS software • Pesticides • Hand lenses • Micro scopes • Entomologist

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KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• describe pest management practices	- Vivipary • Pest management	• Describing pest management practices	
Diseases	• outline the socio -economic importance of diseases • classify diseases according to causal agent • describe signs and symptoms of diseases • describe disease cycle • describe modes of transmission • discuss disease management practices	• Socio-economic importance of diseases. • Classification of diseases - bacterial - fungal - viral • Signs and symptoms • Disease cycle • Modes of transmission • Disease management	• Discussing the socio-economic importance of diseases • Classifying diseases into bacterial, viral and fungal • Examining signs and symptoms of diseases • Discussing disease cycle • Describing modes of transmission • Administering disease management operations	• ICT tools with JAWS software • Specimens of infected plants • Plant pathologist
Safety precautions	• describe safe handling of agro-chemicals • outline safe storage procedures and disposal of agro-chemicals • discuss the effects of agro-chemicals to the environment	• Safety precautions: - handling - storage - disposal	• Discussing safe handling of agro-chemicals • Demonstrating safe handling of agro- chemicals • Researching on the effects of agro-chemicals to the environment	• ICT tools with JAWS software • Safety clothing • Chemical Labels
Sprayer calibration	• calibrate a knapsack sprayer	• Calibration	• Calibrating sprayer	• Knapsack Sprayers • Agro-chemical dealers • ICT tools with JAWS software

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8.6 TOPIC 6: LANDSCAPING AND ORNAMENTAL PLANTS

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Landscaping	• outline the importance of landscaping • identify landscaping features • outline the use of elements in landscaping	• Importance • Features: -plants, (trees, shrubs, hedges, turfs) • Hardscape (stones, pavements, bricks) - buildings - water (Fountains, ponds) • Elements: - line (Bed lines, hardscape lines, path lines, fence lines, tree lines or plant) - form- (informal and formal) - colour (warm, cool) - visual weight (high, low) - texture (fine, coarse)	• Discussing the importance of landscaping • Discussing the usefulness of features in landscaping • Describing the use of elements in landscaping	• Trees • Shrubs • Turfs • Pavements • Bricks • Stones • Water • Landscaping specialist • ICT tools with JAWS software
Principles of landscaping	• explain the principles of landscaping	• Proportion • Order • Repetition • Unity	• Discussing the principles of landscaping	• Trees • Shrubs • Turfs • Pavements • Bricks • Stones • Landscaping specialist • ICT tools with JAWS software
Designing	• carry out needs analysis for specific areas • select appropriate features to	• Needs analysis • Selection of the appropriate features	• Assessing needs • Choosing appropriate features to satisfy the needs of an area	• Trees • Shrubs • Turfs

HORTICULTURE FORMS 5-6 SYLLABUS 2030

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	satisfy the needs of the area • apply different principles and elements of landscaping in designing	• Arrangement of features into elements	• Arranging different features into elements in carrying out landscaping	• Pavements • Bricks • Stones • Landscaping specialist • ICT tools with JAWS software
Ornamental plants	• outline the importance of ornamental plants • discuss the factors that influence choice of ornamental plants • establish ornamental plants • manage established ornamental plants	• Importance • Selection of indigenous and exotic ornamental plants: - trees - shrubs - flowers - herbs - turfs • Plant establishment • Maintenance: - watering - pruning - fertilizing - mowing - shaping - training - spiking	• Discussing the importance of ornamental plants • Describing the factors that influence choice of ornamental plants • Planting ornamental plants • Caring of established plants	•Trees •Shrubs •Turfs •Stones •Landscaping specialist • ICT tools with JAWS software

HORTICULTURE FORMS 5-6 2024-20

8.7 TOPIC 7: FLORICULTURE

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Flower production	• discuss the origins and importance of flowers • describe the agronomic practices followed in flower production • manage a named flower from land preparation to marketing • discuss the post-harvest care of flowers	• Origins and uses • Land preparation • Management practices • Harvesting • Handling and marketing	• Discussing the origins and importance of flowers • Describing the agronomic practices followed in flower production • Establishing and managing flowers • Discussing the post-harvest handling of flowers **NB** Learners should study and produce one plant from the following groups: - Cut flower (roses, chrysanthemum, protease, hypericum) - Pot flower- (African violet, poinsettia)	• ICT tools with JAWS software • Plant samples • Flower samples • Pots

HORTICULTURE FORMS 5&6 2020-2030

8.8 TOPIC 8: VEGETABLE PRODUCTION

SUB TOPIC: VEGETABLE PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Vegetables	• discuss the socio-economic importance of vegetables • describe the classes of vegetables • describe types of vegetable gardens • discuss the significance of vegetable gardens	• Importance • Classification - Use - - scientific - family's edible part • Vegetable gardens – - home garden - floating garden - truck gardens - market gardens - vegetable forcing garden	• Discussing the socio-economic importance of vegetables • Classifying vegetables • Describing types of vegetable gardens • Discussing the significance of vegetable gardens	• ICT tools with JAWS software • Vegetable samples
Vegetable rotation	• explain the principles of vegetable rotation • discuss the benefits of vegetable rotation • design a four-crop cycle vegetable rotation	• Principles • Benefits • Rotational sequence	• Describing the principles of vegetable rotation • Discussing the benefits of vegetable rotation. • Designing a four-crop cycle vegetable rotation	• ICT tools with JAWS software •
Environmental requirements	• describe the soil and climatic requirements of vegetables	• Soil • Climatic requirements	• Discussing soil and climatic requirements of vegetables	• ICT tools with JAWS software • Vegetables
Nursery	• outline factors to consider when selecting a nursery site of a named vegetable • discuss factors to consider when choosing appropriate vegetable	• Establishment - site selection - cultivar selection	• Explaining factors to consider when selecting a nursery site • Discussing factors to consider when choosing appropriate vegetable cultivars	• ICT tools with JAWS software • Nursery site • seeds

HORTICULTURE FORMS 5-6 2024-2030

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	cultivars • manage the nursery	• Management practices	• Carrying out the nursery management practices of a named vegetable	
Vegetable management	• describe the planting and transplanting of a named vegetable • discuss the management of a named vegetable • prepare physical and financial records	• Planting and transplanting • Management: - moisture - fertilizers - weeds - pests - diseases • Record-keeping	• Discussing planting and transplanting of a named vegetable • Discussing management practices of a named vegetable • Growing a named vegetable • Keeping records of vegetables **NB**: One vegetable should be studied and grown from each of the following groups: - legumes: peas/green beans/cowpeas. - leaf: cabbage/lettuce/spinach h - bulbs: onion/garlic - tubers: Irish potato, yams (madhumbe)	• Print and electronic media • ICT tools with JAWS software • Crop inputs
Harvesting	• discuss maturity indices of vegetables • describe methods of harvesting vegetables	• Maturity indices • Methods	• Identifying maturity indices of vegetables • Harvesting mature vegetables	• ICT tools with JAWS software • Print and electronic media
Post-harvest handling and marketing	• describe the storage facilities for vegetables. • discuss the preservation methods and facilities for vegetables • adding value to	• Storage facilities • Preservation facilities • Preservation methods • Value addition	• Describing the storage facilities for vegetables. • Discussing the preservation methods and facilities for vegetables • Processing vegetables	• ICT tools with JAWS software • Vegetables • Samples of processed vegetable products

HORTICULTURE FORMS 5-6 2024-2027

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	vegetables • describe the marketing of vegetables	• Marketing	• Marketing vegetables • Touring a vegetable market	• Print and electronic media

HORTICULTURE FORMS 5-6 2024 2025

FORM SIX

8.9 TOPIC 7: FLORICULTURE

SUB TOPIC: FLOWER PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Flower production	• discuss the origin and the importance of flowers • describe the agronomic practices followed in flower production. • discuss the post-harvest care of flowers • manage a named flower from land preparation to marketing	• Origin and uses • Land preparation • Management practices • Harvesting • Handling and marketing	• Discussing the origin and importance of flowers • Describing the agronomic practices followed in the flower production • Discuss the post-harvest care of flowers • Establishing and managing a flower **NB Learners should only study and produce one plant from the following groups:** - Bed flowers- (petunia, sweet pea, marigold) - Indigenous (cacti, elephant ear, aloe)	• ICT tools with JAWS software • Plant samples • Flower samples • Pots

HORTICULTURE FORMS 5-6 2024-2030

8.10 TOPIC 8: VEGETABLE PRODUCTION

SUB TOPIC: VEGETABLE PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Environmental requirements	• describe the soil and climatic requirements of vegetables	• Soil requirements • Climatic requirements	• Discussing soil and climatic requirements of vegetables	• ICT tools with JAWS software • Vegetable samples • Print and electronic media
Nursery	• outline factors to consider when selecting a vegetable nursery site • discuss factors to consider when choosing appropriate vegetable cultivars • establish and manage a vegetable nursery	• Establishment - site selection - cultivar selection • Management practices	• Explaining factors to consider when selecting a nursery site • Discussing factors to consider when choosing appropriate vegetable cultivars • Carrying out the nursery management practices of a named vegetable	• ICT tools with JAWS software • Nursery site • Seeds • Print and electronic media
Vegetable management	• describe the planting and transplanting of a named vegetable • discuss the management of a named vegetable • prepare physical and financial records	• Planting and transplanting • Management: - moisture - fertilizers - weeds - pests - diseases • Record keeping	• Discussing planting and transplanting of a named vegetable • Discussing management practices of a named vegetable • Growing a named vegetable • Keeping records of vegetables **NB: One vegetable should be studied and grown from each of the following groups:** **-roots: irish potato/ beetroot/carrot/sweet potato** **-fruit vegetables: tomatoes/okra/pepper/mharupwa** **-cucurbits: cucumbers, melons, squashes, butternuts, mapudzi/amakhomane,**	• Print and electronic media • ICT tools with JAWS software • Crop inputs • Seeds of selected varieties

HORTICULTURE FORMS 5-6 2024 - 2030

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
			pumpkins/manhanga/amathanga) **-indigenous leaf vegetables: ulude/munyevhe/imbuya/mowa/tsine**	
Harvesting	• discuss maturity indices of vegetables • describe methods of harvesting vegetables • harvest mature vegetables	• Maturity indices • Methods	• Identifying maturity indices of vegetables • Discussing methods of harvesting mature vegetables • Harvesting mature vegetables	• ICT tools with JAWS software • Print and electronic media
Post-harvest handling and marketing	• describe the storage facilities for vegetables • discuss the preservation methods and facilities for vegetables • describe the process of value addition in vegetables • describe the marketing of vegetables • add value to vegetables • market vegetables	• Storage facilities • Preservation facilities • Value addition • Marketing	• Describing the storage facilities for vegetables. • Discussing the preservation methods and facilities for vegetables • Processing vegetables • Marketing vegetables • Touring a vegetable market	• ICT tools with JAWS software • Samples of processed vegetable products • Storage facilities • Print and electronic media

HORTICULTURE FORMS 5 & 6 2024 - 2028

8.11 TOPIC 9: FRUIT PRODUCTION

SUB TOPIC: FRUIT PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Importance	• outline the socio-economic importance of fruit production	• Socio-economic importance	• Discussing socio-economic importance	• ICT tools with JAWS software
Classification	• classify fruits according to their climatic origins	• Origins: - exotic (tropical, sub-tropical, temperate) - indigenous	• Grouping fruits according to their origins	• Fruits • ICT tools with JAWS software
Nursery	• discuss factors influencing choice of a nursery site • discuss factors to consider when choosing appropriate fruit cultivars • establish fruit tree nursery • carry out management practices in a nursery	• Site selection • Cultivar selection • Establishment • Management	• Discussing factors affecting choice of a nursery site • Discussing factors to consider when choosing appropriate fruit cultivars • Preparing fruit tree nursery • Managing a nursery	• Trays • Pots • Pockets • Fertilizers • Propagation medium • ICT tools with JAWS software • Planting material
Propagation	• use appropriate propagation method	• Methods: - Seed - Cuttings - Grafting - Layering - Budding - Tissue culture - Suckers	• Selecting appropriate propagation methods • Propagating fruit trees	• ICT tools with JAWS software • Seeds • Cuttings • Layering materials • Budding equipment • Rooting hormones • Plants • Suckers
Orchard establishment	• explain the factors which influence site selection • describe orchard land preparation • explain the usefulness of different planting patterns	• Site selection • Land preparation • Planting patterns • Orchard pegging • Planting holes • Planting	• Discussing the factors influencing site selection • Preparing land for an orchard • Discussing the usefulness of different planting patterns • Laying out planting stations	• ICT tools with JAWS software • Fruit trees • Seedlings • Planting boards • Pegs • Lines

HORTICULTURE FORMS 5-6 2025 - 2030

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• peg planting stations • prepare a planting hole • plant fruit trees		• Digging out planting holes • Planting fruit trees	• Digging tools • Organic matter • Fertilizers
Orchard management	• construct conservation structures in an orchard • carry out appropriate management practices in an orchard	• Conservation structures: - basins - ridges • Management practices: - watering - fertilizer application - stacking - pruning - training - weed control - disease control pest control	• Constructing conservation structures • Engaging in appropriate management practices: - watering fruit trees - fertilizing fruit trees - staking and training fruit trees - pruning - controlling weeds, pests and diseases	• Fertilizers • Pesticides • Fungicides • ICT tools with JAWS software • Knapsack sprayers
Harvesting	• identify harvesting indices • harvest fruits using • appropriate harvesting method	• Harvesting Indices • Timing • Methods	• Determining harvesting indices • Harvesting • Touring orchards	• Fruit tress • Harvesting equipment • ICT tools with JAWS software
Post-harvest handling and marketing	• grade the fruits according to set standards • package the graded fruits • process fruits for marketing • determine the price of the produce • market the fruits • prepare production and financial records	• Handling and Marketing: - grading - packaging - weighing - ripening - value addition - storage - transportation	• Grading • Packing • Processing • Pricing • Selling • Compiling production and financial records • Touring fruit processing plants **NB: NB: One fruit crop should be studied and grown from each of the following groups:** **Group A** **Tropical – Mango, banana,**	• ICT tools with JAWS software • Packaging material • Weighing equipment • Ripening hormones

HORTICULTURE FORMS 5-6 2024-2027

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
			avocado, guava **Group B** **<u>Subtropical-</u> Citrus, granadilla, grapes** **Group C** **<u>Temperate</u>** **- Apples, peaches, plums** **Group D** **<u>Indigenous</u>-** **Mawuyu/umkhomo/baobab, mapfura/amaganu, mutohwe/uxakuxaku, nyii/mnyi, masawu, mukute, muzhanje**	

HORTICULTURE FORMS 5 - 2024-2028

8.12 TOPIC 10: HERB AND SPICE PRODUCTION

SUB TOPIC: HERB PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Herb production	• outline the origins and uses of herbs • select appropriate production system • describe land preparation • discuss the management of herbs • harvest, handle and process herbs • market herbs • keep records	• Origins and uses of herbs • Production system - open field - shade • Land preparation • Management practices • Harvesting • Post-harvesting handling and marketing	• Discussing the origins and uses of herbs • Choosing appropriate production system • Preparing land for the production of selected herbs • Establishing and managing the production of herbs • Harvesting, handling and processing herbs • marketing herbs • preparing records **NB Learners should study and grow at least 3 herbs from the following: aloe, mint, rosemary, lavender, wormwood, sweet basil, lemon grass, thyme, cinnamon**	• ICT tools • Herbalist • Herbs

HORTICULTURE FORMS 5 - 6 2024-2030

SUB TOPIC: SPICE PRODUCTION

KEY CONCEPT	LEARNING OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
Spice production	• outline the origin and uses of spices • select appropriate production system • describe land preparation • discuss the management of spices • post-harvest, handle and process spices • market spices • keep records	• Origins and uses of spices • Production system - open field - shade • Land preparation • Management practices • Harvesting • Post-harvesting handling and marketing	• Discussing the origins and uses of spices • Choosing appropriate production system • Preparing land for the production of selected spices • Establishing and managing the production of spices • Harvesting, handling and processing the spices • Marketing spices • Preparing records **NB Leaners should study and grow at least 3 spices from the following: ginger, garlic, chillies, green pepper, coriander, mustard, paprika, sesame, onions**	• ICT tools • Herbalist • Spices • ICT Tools with JAWS software

HORTICULTURE FORMS 5 & 6 2024 - 2030

9.0 ASSESSMENT

The Heritage-based Horticulture syllabus learning area for form 5 and 6 shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners at secondary school level. Arrangements, accommodations and modifications shall be visible to enable candidates with special needs to access assessments.

This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specification grid.

9.1 Assessment Objectives

By the end of the Heritage-based Horticulture Syllabus learning area for form 5 and 6 learners will be assessed on their ability to:

9.1.1 demonstrate an understanding of the importance of horticulture in socio-economic development
9.1.2 apply scientific principles in horticultural production evaluate resources necessary for horticulture production
9.1.3 demonstrate an understanding of plant protection principles
9.1.4 demonstrate the sustainable utilization of local resources
9.1.5 efficiently produce and market horticultural products
9.1.6 add value to horticultural products
9.1.7 design and carry out research work on horticulture production for economic development of the nation
9.1.8 prepare and implement a sustainable horticultural project proposal

9.2 Assessment Model

Assessment of learners at Upper Secondary School Level for Heritage-based Horticulture Syllabus shall be both Continuous and Summative as illustrated in Figure 1. School Based Continuous Assessment shall include recorded activities from the School Based Projects done by the learners. The mark shall be included on learners' end of term and year reports. Summative assessment at school level shall include terminal examinations which are at the end of the term and year.

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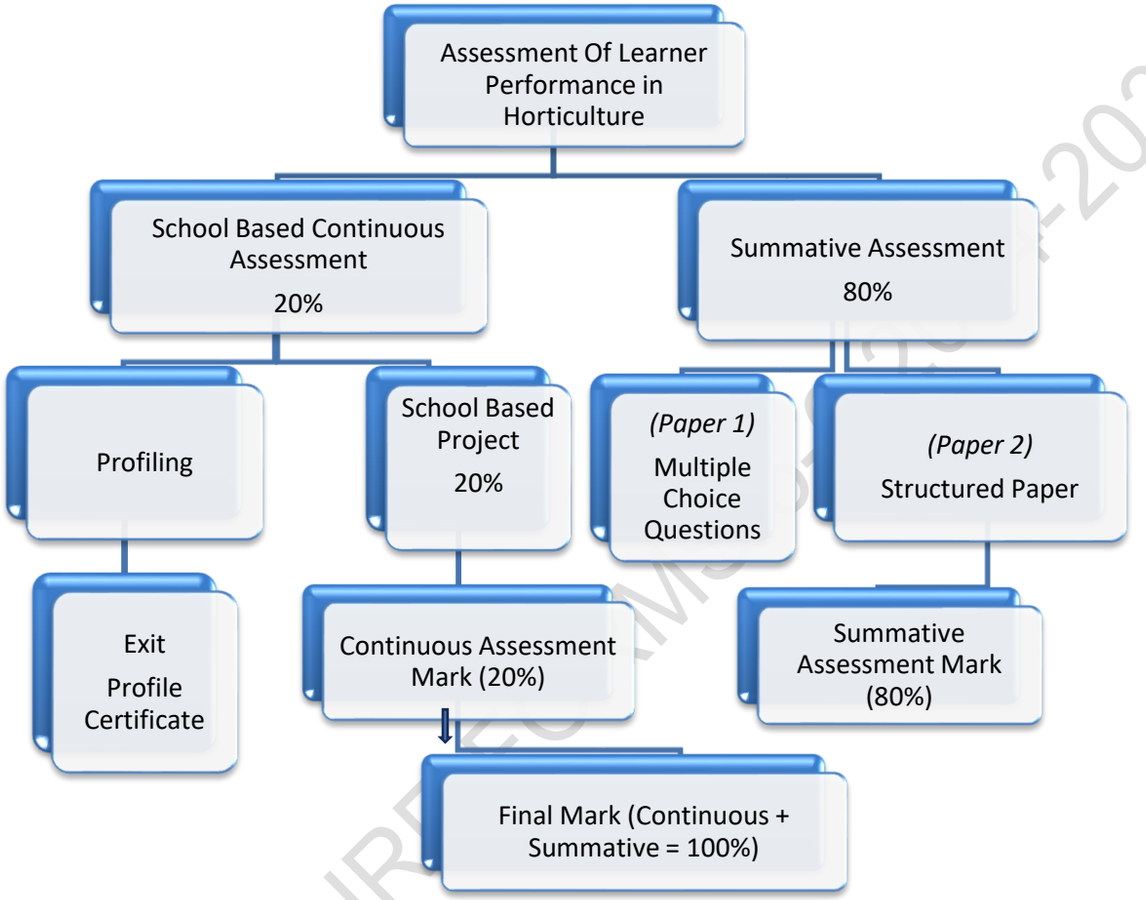


Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established. Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

9.3 Scheme of Assessment

The Assessment Model shows that learners shall be assessed using both School Based Continuous Assessment and Summative Assessment for both School and ZIMSEC assessments.

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The table shows the Scheme of Assessment where 20% is allocated to School Based Continuous Assessment and 80% to School or ZIMSEC Summative Assessment.

FORM OF ASSESSMENT	WEIGHTING
School Based Continuous Assessment	**20%**
Summative Assessment	**80%**
Total	**100%**

9.3.1 Description of School Based Continuous Assessment

Learners shall do one school-based project per form which contributes to 20% of the end of year final mark. The end of year summative assessment shall then contribute 80%. However, for ZIMSEC public examinations, two (2) school-based projects shall be considered as School Based Continuous Assessment at form 5 and 6. The two School Based Projects shall include those done during form 5 and 6 sessions. Each will contribute 10%.

9.3.1.1: School – Based Project Continuous Assessment Scheme

The Table given below shows the Learning and Assessment Scheme for the School Based Project.

Project Execution Stages	Project Stage Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		**50**

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9.3.2 Description of the ZIMSEC Summative Assessment

ZIMSEC Summative Assessment shall be a public examination at Form 6. The examination shall consist of four (4) papers of different weighting.

SUMMATIVE ASSESSMENT 80%

Learners are required to sit for papers 1 to 4.

PAPER DESCRIPTION	DURATION	MARKS	WEIGHTING
Paper 1	1 hour	40	15%
Paper 2	2 hours 30mins	100	40%
Paper 3	2 hours	40	10%
Paper 4	4 terms	100	15%

PAPER 1
Consists of multiple-choice questions from the whole syllabus. Candidates will be required to answer all 40 questions.
Paper total 40

PAPER 2
This is a structured free response paper which has 2 sections namely A and B. Both sections are set from any part of the syllabus.

SECTION A
Candidates will be required to answer all questions in this section. Six questions will be set. Each question is carries10 marks
Section total 60 marks

SECTION B
Essay type questions will be set from any part of the syllabus. Four questions will be set and candidates will be required to answer any 2 questions. Each question carries 20 marks.
Total for section 40
Paper total 100

PAPER 3
A practical examination will be set from any part of the syllabus. The paper will be based on experiments, investigations, observations and calculations. Full instructions will be given where unfamiliar material or techniques will be required. Two compulsory questions are set. Each question carries 20 marks.
Paper total 40

PAPER 4
Candidates will be required to carry out an experimental or a survey project. Candidates will design and carryout the project work on any part of the syllabus. The research project must emphasize both theoretical and practical aspects of Horticulture.

A project report of 2 500 to 3000 words should be prepared and submitted by candidates.
Paper total 100marks

9.5 SCHEME OF ASSESSMENT

HORTICULTURE FORMS 5-6 2024-2030

Skills Specification Grid

ASSESSMENT SKILL	PAPER 1	PAPER 2	PAPER 3	PAPER 4
Knowledge with understanding	50	40	15	10
Application of knowledge	30	40	35	40
Experimental skills	20	20	50	50
Total	**100**	**100**	**100**	**100**

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10.0 Appendix

The following equipment and materials should be available to successfully implement this learning area;

- Controlled environment
  - Green houses
  - Incubators
  - Growth room
- Driers:
  - Air driers
  - Oven driers
- E-learning Solutions software
- Computers
- Land for practical
- Gardening tools
- Cold frames
- Shed house
- Irrigation equipment
- Sprayers
- Protective clothing

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HORTICULTURE FORMS 5-6 2024-2030